

"We Still Use Spreadsheets." Understanding Business Decision-Makers' Perceptions and Barriers to AI Analytics

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Abstract

Although artificial intelligence (AI) systems are expected to support business decision-making, their adoption among non-technical professionals, for instance, in business contexts, remains limited. Prior HCI research in business contexts has mainly explored how technical users provide support to decision-makers, overlooking the challenges of actual business decision-makers themselves. In this work, we present an empirical study surveying 65 participants from business domains and interviewing 18 of them to evaluate their perceptions on analytics and/or AI usage. Our findings reveal that business decision-makers, such as sales professionals, do incorporate data analysis for informed decision-making, which they either delegate to analysts or perform by themselves. Most salespeople still rely on traditional tools, with minimal use of data-driven and AI-based analytics due to adoption barriers that range from skill gaps to an understanding of complex statistical models or tools. These findings inform the need for designing human-centered AI systems that better support business users' decision-making.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Social and professional topics** → *Computer supported cooperative work*; Socio-technical systems.

Keywords

Business Analytics, Business Professionals, Non-Technical Users, Salespeople

ACM Reference Format:

Muhammad Raees, Ioanna Lykourantzou, Vassilis-Javed Khan, and Konstantinos Papangelis. 2026. "We Still Use Spreadsheets." Understanding Business Decision-Makers' Perceptions and Barriers to AI Analytics. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3772363.3798820>



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798820>

1 Introduction

Making informed decisions is important for *business professionals*¹ at various corporate levels [9, 13, 29, 33]. For instance, sales professionals arguably make important business decisions by analyzing customer data (relationships or market insights), transactions (sales or costs), or business operations [1, 9, 20]. Others, such as marketing and operational managers, use data insights for optimizing their execution strategies and enhancing the efficiency of their campaigns [12, 20]. Recent advances, such as the proliferation of Artificial Intelligence (AI) and Machine Learning (ML) tools, are positioned to impact the current work practices of such decision-makers across industries [13, 17, 29]. Despite the growing adoption of AI/ML, limited HCI studies in *business contexts* have explored how business professionals perceive, use, and potentially view the integration of data and AI/ML tools in their contexts for informed decision-making [14, 26, 29].

Business professionals, such as salespeople, do express the need to be able to use AI/ML tools to improve their decisions [1, 10, 14]. Salespeople (managers, executives, and others) are an influential demographic [32] who are at the core of every business, whether a small to medium-sized enterprise or a conglomerate [11, 18, 20, 25]. Therefore, understanding their perceptions and experiences in how AI/ML tools are disrupting their current workflows is important [11]. Existing research examining decision-makers' perceptions of AI/ML analytics [10, 15, 21] has mainly explored the views of technical professionals (e.g., data analysts) who support decision-makers. Studies examining non-technical users often lack exploring perceptions of relevant business professionals [14] or restrict investigations to basic analysis methods [3]. Hence, in this work, we study salespeople's firsthand perceptions of AI/ML analytics (i.e., the tools and technologies they use to meet their current analytics needs). In doing so, we explore the following research question: *What methods and tools do sales professionals use for analytics, and what challenges do they face using AI/ML-based analytics for decision-making?*

We conducted a survey with 65 participants and interviewed 18 sales and business professionals in-depth to evaluate their work practices and the use of advanced AI/ML analytics, investigating the above-mentioned question. Our analysis of survey and interview responses shows that salespeople use various but simple data analysis

¹We use business professionals and business users interchangeably, representing people, such as in sales, marketing, operations, or related fields.

Table 1: A summary of related studies. Prior work has underexplored the contexts of non-technical business professionals, such as sales and marketing professionals, and has focused on understanding their current practices and use of AI in their work.

Study	Demographic	Methods and Main Theme
Alspaugh et al. [4]	Data Analysts	Interviews for exploratory analysis through existing tool usage and potential for automation.
Dimara et al. [10]	Analysts and Decision-Makers	Survey and interviews with analysts and decision-makers for needs for visualization dashboards.
Newburger and Elmquist [21]	Statisticians	Interviews with statisticians on the use of visualizations for statistical inferences.
Gu et al. [15]	Data Analysts	Wizard-of-Oz [8] probe with code notebooks to extract analysts' experiences with AI assistance.
Gathani et al. [14]	Business and Operations Managers	Interviews to extract analysis techniques and decision-making challenges in operations, and identifying potentials for analytics tool usage.

methods and tools (e.g., spreadsheets, CRMs, dashboards) or rely on analysts and their subjective expertise. Our analysis reveals three unique personas described by participants' preferences around the use of advanced tools. These personas also inform why (or why not) AI has (or has not) disrupted their workflows. We also discuss potential considerations for these personas for designing future human-centered tools for them. Our findings provide grounded insights from diverse business professionals through contextual interviews, in contrast to existing studies that rely on a general business audience [3, 10, 14]. Two unique aspects of this study include 1) in-depth interviews with a hard-to-get cohort: business professionals with years of industry experience, and 2) an understanding of AI penetration in actual work settings and business decision-makers' workflows.

2 Background and Related Work

Analytics broadly consists of methods and tools to create insights to answer questions using data analysis [9]. For example, business professionals can apply data models to make sense of customer behavior. Business professionals may employ simple methods or reports (i.e., dashboards) for decisions, or they can use complex analytics (e.g., AI/ML predictive models) that can provide them with insights to adapt/optimize their decisions [1, 18, 32]. However, their expertise and technical capability affect how analytics are used and applied in decision-making [10, 22].

Research has primarily explored studies to understand analytics use by technical experts, such as data analysts [4, 15] or statisticians [21], often overlooking business users (decision-makers) [10, 14, 22]. Table 1 briefly summarizes the related studies exploring analysts' and decision-makers' perspectives. For instance, studies have investigated how analysts or technical users apply analytics [4] to provide support in decision-making, exploring their interactions with visualizations [21], and AI assistance [15]. To differentiate their needs and process workflows, Dimara et al. [10] surveyed and interviewed analysts and decision-makers. Their work [10] focuses on defining (non-technical) visualization tools that can help decision-makers use data analytics more independently, which can help them become self-reliant in decision-making.

In business contexts, Gathani et al. [14] studied the data analysis techniques business professionals apply for decision-making, focusing on people from operations and product management using a sales and marketing scenario. Raees et al. [26] explored a sales context and reported that salespeople face significant challenges while using ML tools for their analysis. Compared to existing studies [10, 14, 26], we elicit salespeople's unique perspectives of

decision-making and the influence of advanced AI/ML analytics methods in their work.

Integrating AI/ML in business users' work has both technical (e.g., skill) and organizational challenges [13, 20]. Business users, like any other non-technical user groups, often lack understanding of complex AI/ML and similar statistical analytics systems and rely on their intuition and experience [25–27]. Hence, understanding the perspectives of business users is highly important for multiple reasons. First, AI is poised to reshape business processes and models across industries, impacting their work [11, 29]. Second, the AI/ML analytics market is estimated to experience 26% yearly growth by 2030 [24, 32], making it increasingly important to enhance their decision-making skills (e.g., in sales, operations, marketing) [9, 20, 28, 29]. Hence, our work focuses on the contexts of business users and their perception of applying AI/ML in their work to enhance their decision-making.

3 Methodology

We conducted this study by running 1) a survey and 2) in-depth interviews. We advertised the user study on Prolific² using focused filtering criteria to recruit target participants having business decision-making roles. On Prolific, we applied filters to work functions (sales), decision-making processes (customers, sales, marketing, supply chain, production), and job titles (vice presidents, directors, managers). The survey focused on understanding participants' business roles and analytics practices. To get deeper insights into respondents' work practices, we conducted detailed interviews. For both cases, we obtained IRB approvals from our institution and followed approved protocols for data collection.

4 Survey Study

4.1 Data Collection

Through a survey (Appendix A), we collected data from 75 respondents with business backgrounds. The survey explained the purpose of the study as *“to gain insights into respondents' decision-making and work practices with data analytics.”* We asked participants about their familiarity with data analytics, including their use of data models (e.g., segment analysis) and AI/ML in their work. Participants also reported their job functions, expertise, and decision-making scenarios. Open feedback was collected about their perception of using advanced tools and methods for performing data analysis and decision-making tasks. In the end, respondents were asked to indicate their interest in a follow-up interview. Survey respondents

²<https://www.prolific.com/>

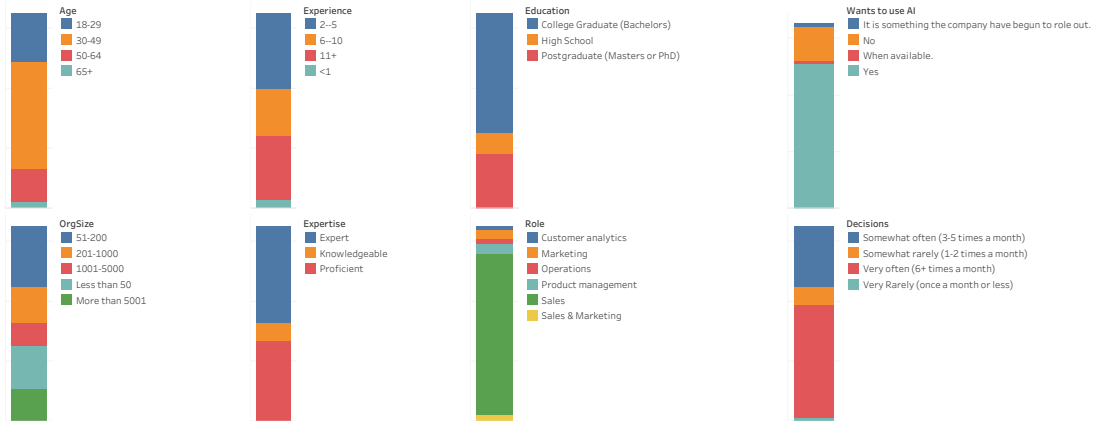


Figure 1: Most survey respondents had college degrees and work experience of more than 6 years. Respondents also reported expertise in their work domains and represented diverse small to large company sizes. Participants had diverse roles in sales and marketing, and were very often involved in decision-making.

received payment at a fair rate of \$12 per hour, with each respondent spending around an average of 20 minutes on the survey.

4.2 Results

To ensure the quality of responses, we embedded two attention check questions in the survey. Around 10 respondents failed the attention check or provided vague responses; hence, 65 respondents were included in the analysis. A brief overview of respondents is provided in Figure 1. Respondents had diverse roles and experience (5+ years mostly) in business decision-making. They reported decision-making tasks ranging from financial tracking to complex behavioral and market research. For instance, respondents commonly used sales performance and financial analysis (profit and loss) on a weekly and monthly basis. In behavioral and market research, respondents mostly conducted segmentation analysis based on customer demographics and buying patterns.

Most respondents reported using Spreadsheets as the main tool for several types of business analysis they conduct. Some also reported using business intelligence (e.g., Tableau³ and Power BI⁴) and Customer Relationship Management (CRM) tools for tracking metrics and customer journeys. Most respondents (82%) expressed the desire to use AI for some of their tasks, while only 55% expressed using it, but only for simple tasks like writing email drafts or sales pitches. However, a few noted that they used tools (e.g., Salesforce⁵) that provide some in-built AI features for analysis. Beyond those, respondents had limited use of advanced tools for their decision-making.

5 Interview Study

For the interview study, we analyzed survey respondents regarding their organizational roles and data-driven decision-making, and recruited 24 participants, 18 of whom completed the full interview study. Incomplete user sessions, happening due to technical or other issues, were excluded. All participants who completed study sessions were employed in sales roles at companies of various sizes. Selected participants held roles like Vice-Presidents (2), Directors (4), sales managers (7 + 2 marketing), and sales advisors (3). They belonged to diverse industries, including education (learning and publishing), manufacturing (products and machines), technology (hardware and software), services (hospitality, health, fitness), and finance (banking and loans). An overview of participants' demographics is shown in Figure 2, and their details are provided in the Appendix B.

5.1 Method and Analysis

The interviews, conducted using a semi-structured approach, focused on participants' work and data, tools, and analytics usage in their contexts. The interview involved 14 open-ended questions (Appendix B). Participants received good compensation at \$35 per hour, due to their sales domain expertise. The interview time was around half an hour ($mean=29.9$, $std=2.80$). Interviews were recorded and transcribed. We analyzed the transcripts using thematic analysis [6] and iterative coding for interpretations [31]. One author created the initial coding based on transcript extracts. Clarifications from other authors were sought to enhance the validity and coherence of the coding. We reviewed the final coded extracts and iteratively collated all the relevant coded data and categorized expressed opinions. Finally, we combined themes, extracted overall patterns, and identified any important/significant outliers. Details on the analysis

³<https://www.tableau.com/>

⁴<https://powerbi.microsoft.com/en-us/>

⁵<https://www.salesforce.com>

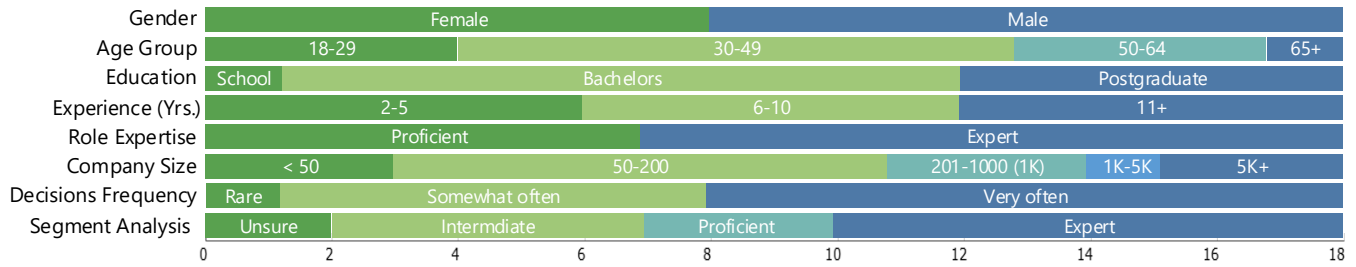


Figure 2: Details of interview participants' demographics. The diversity in demographics, such as age, gender, education, experience (years), companies, decision-making frequency, expertise in their roles, and countries, shows a well-representative sample selected for interviews.

are provided as an Open Science Foundation (OSF) repository⁶ for readers' evaluation.

5.2 Findings

Interviews serve two goals: 1) to validate participants' current practices, and 2) to extract requirements for analytics solutions. Based on interview responses, we extracted three patterns as groups of participants: 1) *generalists*, 2) *formalists*, and 3) *aspirers*. *Generalists* (P5, P6, P9, P12, P17) show subjective dominance and lower aspirations towards advanced analytics, mostly relying on instinct-based and relation-based subjective decisions. Those are mostly involved in supporting sales roles (consultants or representatives) in medium-sized to large companies. *Formalists* (P2, P4, P13, P15, P16) show reliance on analysts and report transition challenges to self-reliance. They acknowledge a lack of control and the inability to change analytics by themselves. They also report communication challenges while relying on analysts. *Aspirers* (P1, P3, P7, P8, P10, P11, P14, P18) want to extend their self-reliance for advanced analysis, enhance objectivity, and expand exploratory analytics. These personas were derived through thematic coding based on recurring patterns identified from interviews. Mainly, three coding dimensions differentiated these personas: 1) their decision-support practices, 2) the complexity of questions and tasks they handle, and 3) the analytics limitations they report. In what follows, these dimensions are explained to distinguish each group.

Decision Support and Practices. *Aspirers* show a high interest in *objective decision-making*, as P3 explains that “we are highly data-oriented and measure [diverse] metrics to identify up-selling opportunities.” However, many participants (11/18) share that “data tells only half the story”, and they optimize their decisions with subjective market insights. Such views are driven by two common patterns: first, for *generalists* it is the dominance of subjective expertise, “Analytics can give you the numbers ... salespeople know how the market works ... tools cannot give you that” (P12), categorized as an *experience-informed* approach. Second, for *others*, combining expertise and analytics gets the best of both worlds, “Analytics may not give a full idea ... human experience can compensate for what tools lack” (P14). Participants also express valuing customer (human) relationships “looking beyond analytics is important in decision-making”

(P2). For example, they rely on intuition or experience, especially at final decision points such as in “negotiating prices/margins” (P14).

Answering Questions. Participants seek to answer diverse questions and measure *Key Performance Indicators (KPIs)* to improve *business outcomes*, but report proficiency challenges (11/18) beyond simple analyses. They conduct simple analyses that can lead to answering their questions, as P15 states, “We look into driving factors used for each marketing channel to see what works [comparing response from channels].” To do simple analysis, they mainly use spreadsheets (15/18), or extract data from CRMs (11/18), e.g., Salesforce, or follow dashboards (9/18). However, most participants (including *aspirers*) struggle with generating insights or conducting complex analysis and rely on external *analysts*. With analysts, they face communication challenges, as P3 points “communicating [domain] problems to data analysts is difficult ... it makes the reliance burdensome.” Despite communication issues, participants praise analysts but acknowledge that their over-dependence is not ideal.

Analytics Limitations. While noting over-reliance on analysts, participants expressed the need to enhance their capabilities. For example, P4 states, “first, we outsourced it ... then we hired analysts ... salespeople cannot do much with current tools”. Hence, most of their current processes are *often manual* (P11, P16), *time-consuming* (P5), *error-prone* (P18) and *inefficient* (P15). The use of AI/ML tools is limited (7/18), even lower for data analysis (4/18), and most usage is to create formulas for the analysis process, using ChatGPT⁷ or co-pilots. Challenges include the lack of skills (13/18), and “steep learning curves” (P4, P6, P11). P1 explains, “first, lack of skill set of people ... then, resources [including training] to implement it in the first place.” P14 also shares, “there is some resistance to change as well with complex tools.” As a result, they aspire to transition slowly, enhancing people's skill set with analytics. In addition, participants think that they have to capture a diverse hierarchy of decision-making when moving away from manual processes, noting “It is much different than mere automation” (P18) and “Explanation of decisions to leadership is easier with manual methods” (P5). Participants also cite *human connection* (e.g., market or customer relationships) and *human touch* as important for them. Such challenges do limit participants' transition to advanced AI/ML analytics.

⁶https://osf.io/f9yta/overview?view_only=7fa7c238b814495b80a77ddd5c80f3

⁷<https://chat.openai.com/>

6 Discussion

Our findings validate that business professionals face challenges with analytics and often rely on subjective insights or data analysts for decision-making. We identify three potential groups of users (*generalists*, *formalists*, and *aspirers*) which inform the nuanced needs of business professionals. Prior work in business analytics categorizes users based on their technical [10] or data skills [14], indicating that broader groups of users have similar expectations from analytics, which we find is not the case for salespeople. Hence, compared to recent work [14, 26], we find business professionals (e.g., *generalists*) show a tendency to use their own subjective insights. *Formalists*, and *aspirers* aim to bring objectivity, but value combining subjective and market knowledge with it. This nuanced yet novel aspect shows that salespeople's tacit knowledge [16], developed through their lived expertise and insights, is a driving factor in decision-making.

Overall, we noted that *generalists* may require simplified data tools where they can adapt (or override) the decisions according to their expertise while examining the data-based support. Literacy and training about data concepts and data-driven decision-making can help *generalists* more than the other persona types. As *formalists* represent a ground between data-driven support and subjective expertise, the decision support that can provide more evidence (e.g., explanations) can be important to reduce their reliance on analysts and be self-reliant. *Formalists* can benefit more from the literacy intervention, which helps them explore the rationale behind decisions. For *aspirers*, easy-to-use advanced tools with exploratory analytics can be more suitable, such as supporting scenario-based testing of decisions. These users may be interested in using step-wise advanced analytics.

We found that participants over-rely on human connections or subjective expertise, under-valuing the use of advanced analytics methods. This aligns with the general notion that non-technical users tend not to fully use new technological tools due to sticking with old mentalities [9, 19]. We hypothesize that such users' limitations are not only related to their knowledge or capabilities of the tools but to their organizational processes as well [13, 30]. For instance, some *generalists* hinted at resistance to change and sticking with subjectivity due to the nature of their roles, along with organizational culture, budgets, and hierarchical issues. Though we did not investigate organizational contexts, our work provides important directions to explore decision management and adoption challenges with advanced AI/ML analytics in those contexts.

Evidently, data analysts play an important role in supporting business users in their analytics needs [10]. However, to enhance business users' ability to conduct data analysis themselves, analytics tools need to be designed by deeply understanding their lived experiences and challenges. For instance, non-technical users may need tools that require low data literacy and fewer execution challenges. Emerging technologies such as language-based analytics [2] can be embedded for guides or customization, as these tools (e.g., co-pilots [7]) greatly execute tasks themselves or facilitate users with step-by-step instructions to execute them.

Limitations. We acknowledge our sample represents only a few disciplines of business professionals. Sales and marketing professionals, nonetheless, are a common role that most organizations have when dealing with direct decision-making on products/services. Therefore, our interpretations elucidate the unique perspectives, some of which can be generalized for other business users like those in procurement/supply-chain roles [23]. We also acknowledge that our findings, as in most qualitative studies, reflect the self-reported experiences of participants.

7 Conclusion and Future Work

Our findings outline that despite increasing usage of AI/ML tools in work contexts, sales professionals still rely on simple analysis or subjective insights to make decisions. While sales professionals have positive views on transitioning towards advanced analytics, they stress the need for the tools to support combining their subjective insights with objective decisions. As advanced analytics sees growth, we identify the need for improving business users' workflows and enhancing their analytics knowledge and skills, while designing human-centered AI tools for them.

This initial work can be extended. First, we anticipate that there could be deeper insights gained from conducting experimental evaluations with specific analytics tools. Introducing an analytics tool to provide users with the ability to perform in-depth (in the wild) analytics techniques (similar to the ones identified by Gathani et al. [14]) to make them self-sufficient in some aspects of their decision-making. Second, the interview study can be expanded with diverse participants from broader business domains. Including participants from broader business domains may elicit more lived experiences and insights in practice, which may be missed in a narrow focus.

Acknowledgments

This work has been supported by Sappi and the Digital Twins Interactions Lab at Rochester Institute of Technology.

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A Survey Questions

- Full Name [Text]
- Age [Text]
- Current or completed education [whichever is higher]
 - School | Bachelor | Master | PhD | Other
- Current or recent job title [Text]
- What role closely aligns with your job
 - Sales | Marketing | Operations | Customer research | Customer analytics | Product management | Other
- Level of expertise in the role
 - Knowledgeable | Proficient | Expert
- How many years of experience do you have in the related role(s) selected above
 - <1 | 2-5 | 6-10 | 11+
- How many employees are there in your organization
 - Less than 50 | 51-200 | 201-1000 | 1001-5000 | More than 5001
- Any other information you would like to share about your role? [Text]
- How often do you make strategic and tactical decisions in your role
 - Very Rarely (once a month or less) | Somewhat rarely (1-2 times a month) | Somewhat often (3-5 times a month) | Very often (6+ times a month)
- Can you explain what type of analysis you perform (customer segment analysis, sales, etc) to make decisions? [Text]
- Have you performed some kind of segmentation analysis (e.g., customer segment analysis, product segment analysis, or similar)? [Text]
- What is your level of expertise in understanding or applying segmentation analysis [Text]
- What type of tools do you use to make this analysis for decision-making (Excel, Sheets, Tableau, Power BI, Internal Company Analytics Apps, Python/code, etc.)? List all [Text]
- Do you use or intend to use predictive analytics (e.g., Machine learning models) for your decision-making?
 - Yes | No
- Do you use Machine Learning or AI in your work or decision-making processes at your work?
- How do you use Machine Learning or AI in your work or decision-making processes?

B Participants and Interview Questions

Table 2: Participants for the interview study. Further details on responses are provided in the OSF project [5].

Sr	Education	Age	Role	Role Job	Experience (yrs)	Expertise	Company Size
P1	Postgraduate	30-49	Sales Director	Sales	2 to 5	Proficient	201-1000
P2	Bachelors	50-64	VP of Sales	Sales	6 to 10	Expert	< 50
P3	Postgraduate	30-49	Sales Director	Sales & Marketing	6 to 10	Expert	51-200
P4	Postgraduate	18-29	Sales Director	Sales	6 to 10	Expert	51-200
P5	Postgraduate	18-29	Sales Manager	Customer analytics	2 to 5	Proficient	5000+
P6	Bachelors	30-49	Sales Consultant	Sales	6 to 10	Expert	5000+
P7	Bachelors	30-49	Sales Manager	Sales	11+	Expert	201-1000
P8	Bachelors	50-64	VP of Sales	Sales	11+	Expert	51-200
P9	Bachelors	18-29	Sales Representative	Sales	2 to 5	Expert	51-200
P10	Postgraduate	50-64	Sales & Marketing Manager	Sales	6 to 10	Expert	51-200
P11	Bachelors	30-49	Sales Director	Sales	11+	Expert	< 50
P12	Postgraduate	30-49	Sales Consultant	Sales	11+	Expert	51-200
P13	Bachelors	30-49	Sales & Marketing Manager	Sales	6 to 10	Proficient	5000+
P14	Bachelors	18-29	Sales Manager	Sales	2 to 5	Proficient	< 50
P15	High School	65+	Sales Manager	Sales	11+	Expert	51-200
P16	Bachelors	30-49	Sales Manager	Sales	2 to 5	Proficient	1001-5000
P17	Bachelors	30-49	Sales Manager	Sales	2 to 5	Proficient	51-200
P18	Bachelors	50-64	Sales Manager	Sales	11+	Proficient	201-1000

Table 3: Interview session questions to understand participants' current practices, challenges, and experiences. We asked questions to participants on their Company and Role, Data and Tools Usage, and Current and Future AI/ML Analytics Usage.

Sr.	Questions
1	Describe your current job.
2	How long have you been in this/similar role?
3	What industry are you in?
4	What are some typical workday tasks you do?
5	How important is data analysis in your decision-making? Follow-up for details.
6	What systems do you use for data analysis? Follow-up for details.
7	Explain the last time you conducted data analysis for your work? Follow-up for details/effort/tools.
8	What challenges do you face using data analysis tools?
9	What role do your colleagues have in the data analysis you conduct?
10	How do you utilize data analytics to make decisions about customers or products? Follow-up for details.
11	How do you conduct/use specific analysis, such as customer segmentation?
12	How do you use AI/ML in your daily work? Follow-up for details.
13	What role does AI/ML have in your data analysis?
14	What challenges/problems do you see using AI/ML analytics in your work? How do you see overcoming those?